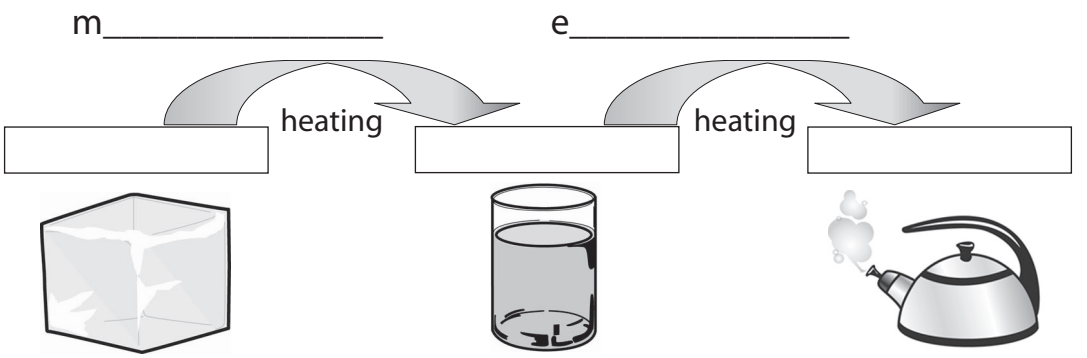
# Heating up

When a substance changes from one state to another, the type of substance does not change. The particles in the substance do not change either. The only things that change are how quickly the particles can move and the amount of space between them. When you heat up a substance, the particles gain **energy**, causing them to move faster and spread out. This changes the substance from a solid to a liquid, or from a liquid to a gas.

☞ **1** Use the words in the word bank to label the diagram below.




**Word bank**
evaporation
liquid
gas
melting
solid

**2** List four everyday examples of substances changing state as they are heated.

_____ _____

_____ _____

**3** Decide whether each of the actions listed in the table below is an example of melting or evaporation.

| Action | Melting or evaporation |
|---|---|
| clothes in a dryer | |
| heating butter in a pan | |
| kettle boiling and producing steam | |

**4** Decide whether each of the following statements is true or false.

**a** When a substance is heated it gains energy. _____

**b** When a substance is heated the particles get bigger. _____

**c** When a substance is heated the particles move further apart. _____

**d** The change of state from a solid to a liquid is called melting. _____


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